

Question Paper

Exam Date & Time: 03-Jan-2026 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - JANUARY 2026

Applications of AI in Government and Business [CSE_4097]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Apply the principles of Explainable AI (XAI) to show how government systems can improve trust and transparency in decision-making. (5)
A)

Course Outcome: "CO1" / Cognitive Level: "Application"

B) Analyze how deep learning addresses the curse of dimensionality and enhances big data analytics in high-dimensional applications. (5)

Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) Analyze how a content-based recommendation system functions by examining the relationships among its key components and how they contribute to generating personalized recommendations. (5)
A)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

B) Apply the concept of model-based algorithms to explain how they are used in collaborative recommendation systems for predicting user preferences. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

C) Analyze the benefits of a hybrid recommendation system. (2)

Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) Apply the concepts of semantic and pragmatic analysis to explain their differences using a suitable example from NLP. (5)
A)

Course Outcome: "CO3" / Cognitive Level: "Application"

B) Analyze how text analytics techniques are utilized in the finance sector to detect and prevent money laundering by examining their role in identifying suspicious patterns and relationships within financial data. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

C) Analyze how deep learning is used for image and voice recognition process. (2)

Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Apply the concepts of descriptive, predictive, and prescriptive analytics and distinguish them (5)

A) through suitable real-world applications.

Course Outcome: "CO4" / Cognitive Level: "Application"

B) Apply the concept of fraud and risk analytics to demonstrate how organizations can detect and prevent financial irregularities. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

C) Analyze why data discovery is essential for AI implementation. (2)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

5) Analyze how the integration of robotics, cognitive insights, and engagement technologies contributes to the formation of an intelligent governance system. (5)

A) Course Outcome: "CO5" / Cognitive Level: "Analysis"

B) Analyze how the Three Vs Framework can be used to evaluate the adoption of cognitive technologies in government programs for enhancing service delivery. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

C) Apply the concept of robotics and demonstrate its role in government applications. (2)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name		Subject	
058-MIT BLR-MAKEUP EXAM-2025-2026		CSE_4097:Applications of AI in Government and Business	
#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name		Subject	
058-MIT BLR-MAKEUP EXAM-2025-2026		CSE_4097:Applications of AI in Government and Business	
#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	6	23(46%)
4	Analysis	8	27(54%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		14	50 (100%)